

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum- 590014, Karnataka.



LAB RECORD on

Big Data Analytics (23CS6PCBDA)

Submitted by

Trisha L Salian (1WA23CS022)

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

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Bull Temple Road, Bangalore 560019

(Affiliated to Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Big Data Analytics” carried out by **Trisha L Salian (1WA23CS022)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

Anitha H K
Assistant Professor
Department of CSE, BMSCE

Dr. Kavitha Sooda
Professor & HOD
Department of CSE, BMSCE

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Github Link: https://github.com/Trishalsalian/BDA_LAB

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
mongosh mongodb
bmscece@bmscece-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~
bmscece@bmscece-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ mongosh
Current Mongosh Log ID: 69f86b097563def5c351e943
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverS
Using MongoDB:      8.0.5
Using Mongosh:       2.4.0
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/sh
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-04T13:47:22.247+05:30: Using the XFS filesystem is strongly recommend
2026-05-04T13:47:25.186+05:30: Access control is not enabled for the database
2026-05-04T13:47:25.186+05:30: For customers running the current memory alloc
2026-05-04T13:47:25.186+05:30: For customers running the current memory alloc
2026-05-04T13:47:25.186+05:30: We suggest setting the contents of sysfsFile t
2026-05-04T13:47:25.186+05:30: We suggest setting swappiness to 0 or 1, as sw
-----

test> use employee
switched to db employee
employee> db.createCollection("emp_info")
employee> db.emp_info.insertMany([{id:1,name:"a",desg:"aa",date:ISODate("2026-01-01"),salary:10000,dept:"ccc"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f86dcd7563def5c351e944'),
    '1': ObjectId('69f86dcd7563def5c351e945'),
    '2': ObjectId('69f86dcd7563def5c351e946')
  }
}
employee> db.emp_info.updateOne({id:1},{ $set:{name:"David",dept:"ECE"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
employee> db.employee.find().pretty()
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'a',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ccc'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'a',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ccc'
  }
]
```

```
bmscscse@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  }
]
employee> db.emp_info.find().sort({salary:1})
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  }
]
employee> db.emp_info.find().sort({salary:-1})
[
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  }
]
employee> db.emp_info.updateOne({id:1},{ $set:{projects:["data"]}})
{ acknowledged: true, insertedId: null, matchedCount: 1, modifiedCount: 1, upsertedCount: 0 }
employee> db.emp_info.updateOne({id:2},{ $set:{projects:["machine learning"]}})
{ acknowledged: true, insertedId: null, matchedCount: 1, modifiedCount: 1, upsertedCount: 0 }
employee> db.emp_info.updateOne({id:3},{ $set:{projects:["BDA"]}})
{ acknowledged: true, insertedId: null, matchedCount: 1, modifiedCount: 1, upsertedCount: 0 }
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
bmscscse@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~
employee> db.emp_info.insertOne({id:4,name:"d",dept:"dd",date:ISODate("2026-01-04"),salary:40000,desg:"ddd",expireAt:new Date(Date.now()+15000)})
{ acknowledged: true, insertedId: ObjectId('69f874177563def5c351e947') }
employee> db.emp_info.createIndex({expireAt:1},{expireAfterSeconds:0})
expireAt_1
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE',
    projects: [ 'data analytics' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb',
    projects: [ 'machine learning' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc',
    projects: [ 'BDA' ]
  },
  {
    _id: ObjectId('69f874177563def5c351e947'),
    id: 4,
    name: 'd',
    dept: 'dd',
    date: ISODate('2026-01-04T00:00:00.000Z'),
    salary: 40000,
    desg: 'ddd',
    expireAt: ISODate('2026-05-04T10:25:42.188Z')
  }
]
employee> █
```


Perform the following DB operations using Cassandra Questions:

```
(3 rows)
cqlsh:emp> insert into emp(id,name,dept,date,salary) values(0,'p','pp','2026-01-04',40000);
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

```
(4 rows)
cqlsh:emp> update emp set dept='xx',name='a',salary=10000 where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	xx	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

```
(4 rows)
cqlsh:emp> delete dept='xx' from emp where id=0;
SyntaxException: line 1:11 no viable alternative at input '=' (delete [dept]=...)
cqlsh:emp> delete dept from emp where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	null	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

```
(4 rows)
cqlsh:emp> delete dept:'xx' from emp where id=0;
SyntaxException: line 1:11 no viable alternative at input ':' (delete [dept]:...)
cqlsh:emp> delete from emp where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

```
(3 rows)
cqlsh:emp> alter table emp add grade set<text>;
cqlsh:emp> select * from emp;
```

id	date	dept	grade	name	salary
----	------	------	-------	------	--------


```

(3 rows)
cqlsh:emp> alter table emp add grade set<text>;
cqlsh:emp> select * from emp;

 id | date                | dept | grade | name | salary
-----+-----+-----+-----+-----+-----
  1 | 2025-12-31 18:30:00.000000+0000 | xx | null | a | 10000
  2 | 2026-01-01 18:30:00.000000+0000 | yy | null | b | 20000
  3 | 2026-01-02 18:30:00.000000+0000 | zz | null | c | 30000

(3 rows)
cqlsh:emp> update emp set grade={'a','b'} where id=1;
cqlsh:emp> update emp set grade={'a','b'} where id=2;
cqlsh:emp> update emp set grade={'a','b','c'} where id=3;
cqlsh:emp> select * from emp;

 id | date                | dept | grade                | name | salary
-----+-----+-----+-----+-----+-----
  1 | 2025-12-31 18:30:00.000000+0000 | xx | {'a', 'b'}          | a | 10000
  2 | 2026-01-01 18:30:00.000000+0000 | yy | {'a', 'b'}          | b | 20000
  3 | 2026-01-02 18:30:00.000000+0000 | zz | {'a', 'b', 'c'}     | c | 30000

(3 rows)
cqlsh:emp> alter table emp add marks list<int>;
cqlsh:emp> update emp set list={1,2,3} where id=0;
InvalidRequest: Error from server: code=2200 [Invalid query] message="Undefined column name list"
cqlsh:emp> select * from emp;

 id | date                | dept | grade                | marks | name | salary
-----+-----+-----+-----+-----+-----+-----
  1 | 2025-12-31 18:30:00.000000+0000 | xx | {'a', 'b'}          | null | a | 10000
  2 | 2026-01-01 18:30:00.000000+0000 | yy | {'a', 'b'}          | null | b | 20000
  3 | 2026-01-02 18:30:00.000000+0000 | zz | {'a', 'b', 'c'}     | null | c | 30000

(3 rows)
cqlsh:emp> update emp set marks={1,2,3} where id=1;
InvalidRequest: Error from server: code=2200 [Invalid query] message="Invalid set literal for marks"
cqlsh:emp> update emp set marks=[1,2,3] where id=1;
cqlsh:emp> update emp set marks=[1,2,3] where id=2;
cqlsh:emp> update emp set marks=[1] where id=3;
cqlsh:emp> select * from emp;

 id | date                | dept | grade                | marks | name | salary
-----+-----+-----+-----+-----+-----+-----
  1 | 2025-12-31 18:30:00.000000+0000 | xx | {'a', 'b'}          | [1, 2, 3] | a | 10000
  2 | 2026-01-01 18:30:00.000000+0000 | yy | {'a', 'b'}          | [1, 2, 3] | b | 20000
  3 | 2026-01-02 18:30:00.000000+0000 | zz | {'a', 'b', 'c'}     | [1] | c | 30000

(3 rows)
cqlsh:emp> select dept as "department" from emp;

 department
-----

```

```
cqlsh:emp> update emp set marks=[1,2,3] where id=2;
cqlsh:emp> update emp set marks=[1] where id=3;
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(3 rows)

```
cqlsh:emp> select dept as "department" from emp;
```

department
xx
yy
zz

(3 rows)

```
cqlsh:emp> select dept as 'department' from emp;
```

SyntaxException: line 1:15 no viable alternative at input 'department' (select dept as ['depart

```
cqlsh:emp> insert into emp(id,name,dept,date,salary) values(0,'p','pp','2026-01-04',40000) usin
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	null	null	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(4 rows)

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	null	null	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(4 rows)

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(3 rows)

```
cqlsh:emp>
```

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

- Create a keyspace by name Library
- Create a column family by name Library-Info with attributes
 - Stud_Id Primary Key,
 - Counter_value of type Counter,
 - Stud_Name, Book-Name, Book-Id,
 - Date_of_issue
- Insert the values into the table in batch
- Display the details of the table created and increase the value of the counter
- Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
- Export the created column to a csv file
- Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh:employee> CREATE KEYSPACE IF NOT EXISTS Library
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh:employee> USE Library;
cqlsh:library> CREATE TABLE IF NOT EXISTS Library_Info (
...   Stud_Id INT PRIMARY KEY,
...   Stud_Name TEXT,
...   Book_Name TEXT,
...   Book_Id TEXT,
...   Date_of_issue DATE
... );
cqlsh:library> CREATE TABLE IF NOT EXISTS Book_Counter (
...   Stud_Id INT,
...   Book_Name TEXT,
...   Counter_value COUNTER,
...   PRIMARY KEY ((Stud_Id), Book_Name)
... );
cqlsh:library> BEGIN BATCH
...   INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
...   VALUES (112, 'Anjali Rao', 'BDA', 'B101', '2024-10-01');
...
...   INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
...   VALUES (113, 'Karthik N', 'AI', 'B102', '2024-11-11');
... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> UPDATE Book_Counter SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Book_Counter WHERE Stud_Id = 112 AND Book_Name = 'BDA';

stud_id | book_name | counter_value
-----+-----+-----
112 | BDA | 4
(1 rows)
```

```
cqlsh:students> DESCRIBE TABLE Students_Info;

CREATE TABLE students.students_info (
  roll_no int PRIMARY KEY,
  dateofjoining timestamp,
  last_exam_percent double,
  studname text
) WITH additional_write_policy = '99p'
AND bloom_filter_fp_chance = 0.01
AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
AND cdc = false
AND comment = ''
AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
AND mentable = 'default'
AND crc_check_chance = 1.0
AND default_time_to_live = 0
AND extensions = {}
AND gc_grace_seconds = 864000
AND max_index_interval = 2048
AND memtable_flush_period_in_ms = 0
AND min_index_interval = 128
AND read_repair = 'BLOCKING'
AND speculative_retry = '99p';
cqlsh:students> BEGIN BATCH
... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
... VALUES (1, 'Asha', '2012-03-12', 79.9);
... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
... VALUES (2, 'Kiran', '2012-03-12', 89.9);
... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
... VALUES (3, 'Shanthi', '2012-03-12', 90.9);
... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
... VALUES (4, 'Smith', '2012-03-12', 67.9);
... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
... VALUES (5, 'Rohan', '2012-03-12', 56.9);
... APPLY BATCH;
cqlsh:students> SELECT * FROM Students_Info;

roll_no | dateofjoining | last_exam_percent | studname
-----|-----|-----|-----
5 | 2012-03-11 18:30:00.000000+0000 | 56.9 | Rohan
1 | 2012-03-11 18:30:00.000000+0000 | 79.9 | Asha
2 | 2012-03-11 18:30:00.000000+0000 | 89.9 | Kiran
4 | 2012-03-11 18:30:00.000000+0000 | 67.9 | Smith
3 | 2012-03-11 18:30:00.000000+0000 | 90.9 | Shanthi

(5 rows)
```

```
cqlsh> CREATE KEYSPACE Students WITH REPLICATION =
... {'class': 'SimpleStrategy', 'replication_factor': '1'};
cqlsh>
cqlsh> USE Students;
cqlsh:students> DESCRIBE KEYSPACES;

companies | library | products | system | system_traces
company | pro | productsss | system_auth | system_views
employe | prod | productsss | system_distributed | system_virtual_schema
employee | productname | students | system_schema

cqlsh:students> CREATE TABLE Students_Info (
... Roll_No int PRIMARY KEY,
... StudName text,
... DateOfJoining timestamp,
... last_exam_Percent double
... );
cqlsh:students> SELECT * FROM system.schema_keyspaces;
InvalidRequest: Error from server: code=2200 [Invalid query] message="table schema_keyspaces does not exist"
cqlsh:students>
cqlsh:students> SELECT * FROM system_schema.keyspaces;

keyspace_name | durable_writes | replication
-----|-----|-----
companies | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_auth | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_schema | True | {'class': 'org.apache.cassandra.locator.LocalStrategy'}
library | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
products | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_distributed | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '3'}
system | True | {'class': 'org.apache.cassandra.locator.LocalStrategy'}
productsss | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
prod | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
pro | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
system_traces | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '2'}
students | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
company | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
employee | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
productname | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
employe | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
productss | True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}

(17 rows)
cqlsh:students> DESCRIBE TABLES;

students_info
```



```

cqlsh:students> SELECT * FROM Students_Info WHERE Roll_No IN (1,2,3);

roll_no | dateofjoining | last_exam_percent | studname
-----+-----+-----+-----
1 | 2012-03-11 18:30:00.000000+0000 | 79.9 | Asha
2 | 2012-03-11 18:30:00.000000+0000 | 89.9 | Kiran
3 | 2012-03-11 18:30:00.000000+0000 | 90.9 | Shanthi

(3 rows)
cqlsh:students> CREATE INDEX ON Students_Info (StudName);
cqlsh:students> SELECT * FROM Students_Info WHERE StudName = 'Asha';

roll_no | dateofjoining | last_exam_percent | studname
-----+-----+-----+-----
1 | 2012-03-11 18:30:00.000000+0000 | 79.9 | Asha

(1 rows)
cqlsh:students> SELECT Roll_No, StudName FROM Students_Info LIMIT 2;

roll_no | studname
-----+-----
5 | Rohan
1 | Asha

(2 rows)
cqlsh:students> SELECT Roll_No AS USN FROM Students_Info;

usn
----
5
1
2
4
3

(5 rows)
cqlsh:students> UPDATE Students_Info
... SET StudName = 'David Sheen'
... WHERE Roll_No = 2;
cqlsh:students> UPDATE Students_Info SET Roll_No = 6 WHERE Roll_No = 3; -- ✗ ERROR!
InvalidRequest: Error from server: code=2200 [invalid query] message="PRIMARY KEY part roll_no found in SET part"

cqlsh:students> DELETE Last_Exam_Percent FROM Students_Info WHERE Roll_No = 2;
cqlsh:students> DELETE FROM Students_Info WHERE Roll_No = 2;
cqlsh:students> ALTER TABLE Students_Info ADD hobbies SET<text>;
cqlsh:students> ALTER TABLE Students_Info ADD languages LIST<text>;
cqlsh:students> UPDATE Students_Info
... SET hobbies = hobbies + {'Chess', 'Table Tennis'}
... WHERE Roll_No = 1;
cqlsh:students> CREATE TABLE library_book (
... counter_value counter,
... book_name text,
... stud_name text,
... PRIMARY KEY(book_name, stud_name)
... );
cqlsh:students> UPDATE library_book
... SET counter_value = counter_value + 1
... WHERE book_name = 'Big Data Analytics' AND stud_name = 'Jeet';
cqlsh:students> CREATE TABLE userlogin (
... userid int PRIMARY KEY,
... password text
... );
cqlsh:students> INSERT INTO userlogin (userid, password)
... VALUES (1, 'infy') USING TTL 30;
cqlsh:students> SELECT TTL(password) FROM userlogin WHERE userid = 1;

ttl(password)
-----
20

(1 rows)
cqlsh:students> COPY Students_Info TO '/home/bmscece/Desktop/Student_Info.csv';
Using 16 child processes

Starting copy of students.students_info with columns [roll_no, dateofjoining, hobbies, languages, last_exam_percent, studname].
Processed: 4 rows; Rate: 38 rows/s; Avg. rate: 38 rows/s
4 rows exported to 1 files in 0.124 seconds.
cqlsh:students> COPY Students_Info FROM '/home/bmscece/Desktop/Student_Info.csv';
Using 16 child processes

Starting copy of students.students_info with columns [roll_no, dateofjoining, hobbies, languages, last_exam_percent, studname].
Processed: 4 rows; Rate: 7 rows/s; Avg. rate: 11 rows/s
4 rows imported from 1 files in 0.377 seconds (0 skipped).
cqlsh:students> COPY person (id, fname, lname) FROM STDIN;
Column family person not found
cqlsh:students> COPY Students_Info TO STDOUT;
5,2012-03-11 18:30:00.000+0000,,56.9,Rohan
1,2012-03-11 18:30:00.000+0000,{'Chess', 'Table Tennis'},79.9,Asha
4,2012-03-11 18:30:00.000+0000,,67.9,Smith
3,2012-03-11 18:30:00.000+0000,,90.9,Shanthi
cqlsh:students>

```

LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ cd ./Desktop/
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab05
```

```
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Hadoop
ls: '/Hadoop': No such file or directory
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
```

```
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ touch test.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano test.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put ./text.txt /Lab05/text.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 1 items
-rw-r--r-- 1 hadoop supergroup 19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/text.txt
Hello
How are you?
```

```
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /Lab05
Found 2 items
-rw-r--r-- 1 hadoop supergroup 15 2024-05-13 14:40 /Lab05/test.txt
-rw-r--r-- 1 hadoop supergroup 19 2024-05-13 14:33 /Lab05/text.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05 /text.txt /Lab05 /test.txt ../
Downloads/Merged.txt
getmerge: '/text.txt': No such file or directory
getmerge: '/test.txt': No such file or directory
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /Lab05/text.txt /Lab05/test.txt ../Do
wnloads/Merged.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /Lab05
# file: /Lab05
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

```
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -put /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/Welcome.txt /abc/WC.txt
copyFromLocal: '/abc/WC.txt': File exists
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -get /abc/WC.txt /home/hadoop/Downloads/WMC.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -getmerge /abc/ /home/hadoop/Desktop/Merge.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -getfacl /abc/
# file: /abc
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /abc/WC.txt
hello world
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -mv /abc /FFF
hadoop fs -ls /FFF
Found 3 items
-rw-r--r-- 1 hadoop supergroup 12 2025-04-15 14:53 /FFF/WC.txt
-rw-r--r-- 1 hadoop supergroup 12 2024-05-14 14:35 /FFF/file.txt
-rw-r--r-- 1 hadoop supergroup 12 2024-05-14 14:38 /FFF/file_cp_local.txt
hadoop@bmscsecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cp /CSE/ /LLL
```



```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/test.txt ../Documents
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -copyToLocal /Lab05/test.txt ../Documents
```

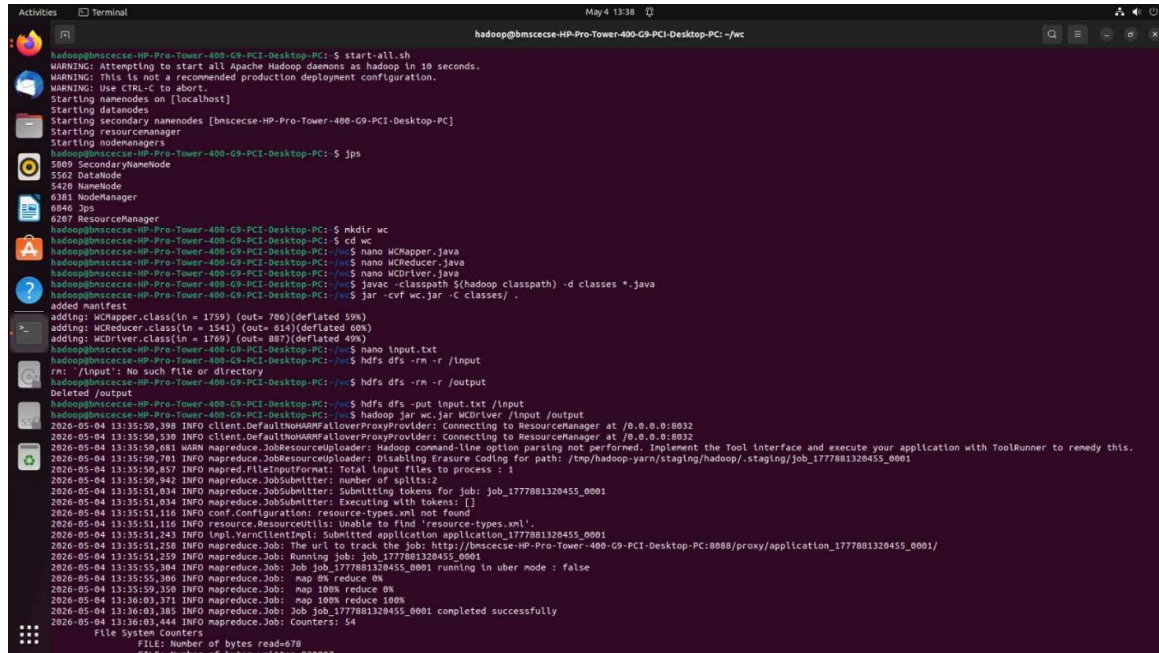
```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cat /Lab05/test.txt
Hello
How are you?
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mv /Lab05 /test_Lab05
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r-- 1 hadoop supergroup 15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r-- 1 hadoop supergroup 19 2024-05-13 14:33 /test_Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -cp /test_Lab05/ /Lab05
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /Lab05
Found 2 items
-rw-r--r-- 1 hadoop supergroup 15 2024-05-13 14:51 /Lab05/test.txt
-rw-r--r-- 1 hadoop supergroup 19 2024-05-13 14:51 /Lab05/text.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -ls /test_Lab05
Found 2 items
-rw-r--r-- 1 hadoop supergroup 15 2024-05-13 14:40 /test_Lab05/test.txt
-rw-r--r-- 1 hadoop supergroup 19 2024-05-13 14:33 /test_Lab05/text.txt
```

LABORATORY PROGRAM – 5

Implement Wordcount program on Hadoop framework

CODE, COMMAND WITH OUTPUT



```
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ jps
5889 SecondaryNameNode
5562 DataNode
5410 NameNode
6381 NodeManager
6846 Jps
6297 ResourceManager
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ mkdir wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ cd wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ nano WCMapper.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ nano WCRReducer.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ nano WCDriver.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ javac -classpath $(hadoop classpath) -d classes *.java
added manifest
adding: WCMapper.class(in = 1759) (out= 786)(deflated 59%)
adding: WCRReducer.class(in = 1241) (out= 614)(deflated 60%)
adding: WCDriver.class(in = 1769) (out= 887)(deflated 49%)
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ nano input.txt
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ hdfs dfs -rm -r /Input
rm: /Input: No such file or directory
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ hdfs dfs -rm -r /output
Deleted /output
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc $ hdfs dfs -put input.txt /input
2026-05-04 13:35:50,398 INFO client.DefaultHadoopHwallerProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 13:35:50,530 INFO client.DefaultHadoopHwallerProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 13:35:50,681 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-04 13:35:50,701 INFO mapreduce.JobResourceUploader: Disabling Erasure coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1777881320455_0001
2026-05-04 13:35:50,857 INFO mapred.FileInputFormat: Total input files to process : 1
2026-05-04 13:35:50,942 INFO mapreduce.JobSubmitter: number of splits:2
2026-05-04 13:35:51,034 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777881320455_0001
2026-05-04 13:35:51,034 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-04 13:35:51,116 INFO conf.Configuration: resource-types.xml not found
2026-05-04 13:35:51,116 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-04 13:35:51,243 INFO impl.YarnClientImpl: Submitted application application_1777881320455_0001
2026-05-04 13:35:51,258 INFO mapreduce.Job: The url to track the job: http://bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:8088/proxy/application_1777881320455_0001/
2026-05-04 13:35:51,259 INFO mapreduce.Job: Running job: job_1777881320455_0001
2026-05-04 13:35:55,304 INFO mapreduce.Job: Job job_1777881320455_0001 running in uber mode : false
2026-05-04 13:35:55,306 INFO mapreduce.Job: map 0% reduce 0%
2026-05-04 13:35:59,350 INFO mapreduce.Job: map 100% reduce 0%
2026-05-04 13:36:03,371 INFO mapreduce.Job: map 100% reduce 100%
2026-05-04 13:36:03,385 INFO mapreduce.Job: Job job_1777881320455_0001 completed successfully
2026-05-04 13:36:03,444 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=678
FILE: Number of bytes written=10007
```

Driver Code

```
// Importing libraries import java.io.IOException;
import org.apache.hadoop.conf.Configured; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.FileInputFormat; import
org.apache.hadoop.mapred.FileOutputFormat; import
org.apache.hadoop.mapred.JobClient; import
org.apache.hadoop.mapred.JobConf; import
org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }
        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCRReducer.class);
```

```

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }
    // Main Method
    public static void main(String[] args)
throws Exception {
        int exitCode =
        ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}

```

Mapper Code

```

// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word :
line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}

```

Reducer Code

```

// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key,
Iterator<IntWritable> values,
OutputCollector<Text, IntWritable> output,
Reporter reporter) throws IOException {
        int count = 0;
    }
}

```

```
// Counting the frequency of each word
while (values.hasNext()) {
    count += values.next().get();
}

output.collect(key, new IntWritable(count));
}
```

```
hadoop@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$ hdfs dfs -cat /output/part-00000
15
a      2
b      2
e      3
f      1
h      7
i      7
j      1
l      1
n      1
o      12
r      8
s      6
t      2
u      5
w      5
y      6
hadoop@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/wc$
```


LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all>

- Create a MapReduce program to find average temperature for each year from NCD data set.
- find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A

```
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc/meanmax
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc/meanmax$ cd -
bash: cd: QLDPM not set
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc/meanmax$ cd
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ cd
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 5420. Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 5562. Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC]
bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: secondarynamenode is running as process 5809. Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 6207. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 6381. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ jps
5809 SecondaryNameNode
14937 Jps
5562 DataNode
5420 NameNode
6381 NodeManager
6207 ResourceManager
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ mkdir meanmax
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ cd meanmax
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ nano MeanMaxDriver.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ nano MeanMaxMapper.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ nano MeanMaxReducer.java
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ mkdir classes
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ javac -classpath $(hadoop classpath) -d classes *.java
Note: MeanMaxDriver.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ jar -cvf meanmax.jar -C classes/ .
added manifest
```

```
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc/meanmax
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/wc/meanmax$ wget https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1902.gz
--2026-05-04 14:31:11-- https://raw.githubusercontent.com/tomwhite/hadoop-book/master/input/ncdc/all/1902.gz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199.109.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 74105 (72K) [application/octet-stream]
Saving to: '1902.gz'
1902.gz
100%[=====] 72.37K --.-KB/s in 0.05s
2026-05-04 14:31:12 (1.53 MB/s) - '1902.gz' saved [74105/74105]
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ gunzip -c 1901.gz > weather.txt
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ hdfs dfs -rm -r /meaninput
Deleted /meaninput
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ hdfs dfs -rm -r /meanoutput
Deleted /meanoutput
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ hdfs dfs -mkdir /meaninput
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ hdfs dfs -put weather.txt /meaninput
hadoop@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~/meanmax$ hadoop jar meanmax.jar MeanMaxDriver /meaninput /meanoutput
2026-05-04 14:32:39,823 INFO client.DefaultHARMFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-04 14:32:39,962 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool Interface and execute your ap
2026-05-04 14:32:39,987 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1777881320455_0005
2026-05-04 14:32:40,151 INFO Input.FileInputFormat: Total Input files to process : 1
2026-05-04 14:32:40,244 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-04 14:32:40,335 INFO mapreduce.JobSubmitter: number of splits:1
```

Driver Code

package temp;

```
import org.apache.hadoop.conf.Configuration; import
org.apache.hadoop.fs.Path;
```

```

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }
        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Mapper Code

```

package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.LongWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();
    }
}

```

```

        // Extract year from fixed position
String year = line.substring(15, 19);    int
temperature;

        // Determine if there's a '+' sign    if (line.charAt(87)
== '+') {    temperature =
Integer.parseInt(line.substring(88, 92));
    } else {
        temperature = Integer.parseInt(line.substring(87, 92));
    }

    // Quality check character
String quality = line.substring(92, 93);

    // Only emit if data is valid
if (temperature != MISSING && quality.matches("[01459]")) {
    context.write(new Text(year), new IntWritable(temperature));
}
}
}

```

Reducer Code

```

package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
context.write(key, new IntWritable(average));    }
        }
    }
}

```

CODE, COMMAND WITH OUTPUT – B

Driver Code

```

package meanmax;

```

```

import org.apache.hadoop.conf.Configuration; import
org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Mean and Max Temperature");

        job.setJarByClass(MeanMaxDriver.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.setMapperClass(MeanMaxMapper.class);
        job.setReducerClass(MeanMaxReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Mapper Code

```

package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();
    }
}

```

```

        // Extract month from positions 19-20
String month = line.substring(19, 21);    int
temperature;

        // Extract temperature considering optional '+'
if (line.charAt(87) == '+') {
    temperature = Integer.parseInt(line.substring(88, 92));
} else {
    temperature = Integer.parseInt(line.substring(87, 92));
}

        // Quality check
String quality = line.substring(92, 93);

        if (temperature != MISSING && quality.matches("[01459]")) {
context.write(new Text(month), new IntWritable(temperature));
        }
    }
}

```

Reducer Code

```

package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context) throws IOException, InterruptedException {

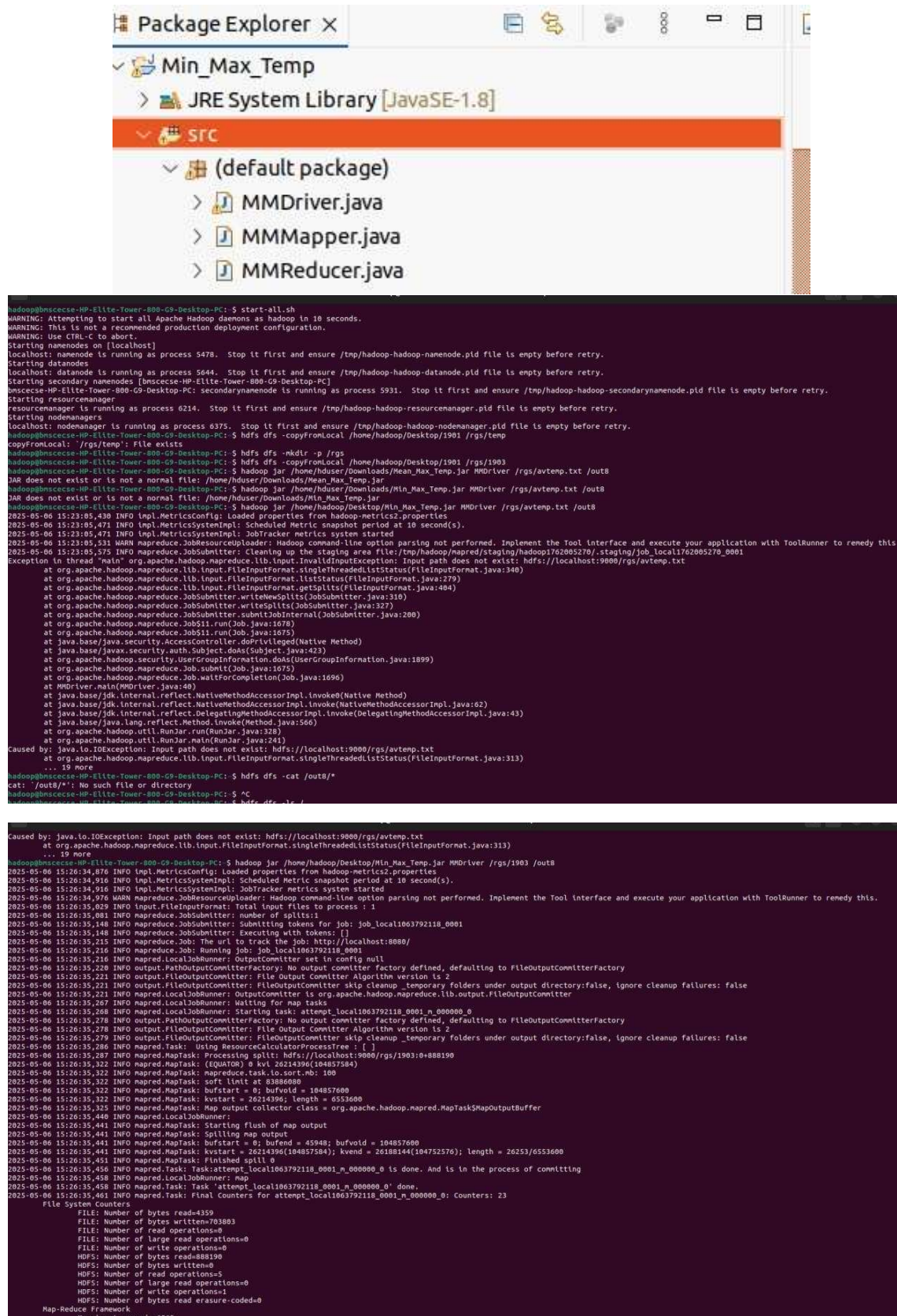
        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
int temp = value.get();
sumTemp += temp;        count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
context.write(key, new Text(result));
        }
    }
}

```

```

hadoop@bnscece-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/neannax$ hdfs dfs -cat /meanoutput/part-r-00000
01      4
02      1
03      6
04     34
05     90
06    144
07    184
08    172
09    123
10     73
11     21
12      3
hadoop@bnscece-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/neannax$

```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

CODE, COMMAND WITH OUTPUT

Driver Code (TopNDriver.java)

```

package samples.topn;

import org.apache.hadoop.conf.Configuration; import
org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job; import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }
        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);
    }
}

```

```

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }
        // === Job 2: Top N ===
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    } }

```

Mapper Code (WordCountMapper.java)

```

package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper extends
Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text(); // characters to normalize
    into spaces
    private String tokens = "[_!$%&'\\";

    @Override
    protected void map(Object key, Text value, Context context)
    throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
            .toLowerCase()
            .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}

```

Mapper Code (TopNMapper.java)

```

package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;

```

```

import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper extends Mapper<Object,
Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
    throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1])); //
            emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}

```

Reducer Code (WordCountReducer.java)

```

package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer extends Reducer<Text,
IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
    throws IOException, InterruptedException {

        int sum = 0;    for
        (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}

```

Reducer Code (TopNReducer.java)

```

package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List; import
java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

```

```

public class TopNReducer extends
Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
private TreeMap<Integer, List<String>>> countMap =
    new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }
    @Override
    protected void cleanup(Context context)
throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();    int seen = 0;
        for (Map.Entry<Integer, List<String>>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();    for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;
                if (seen == 10) break;
            }
            if (seen == 10) break;
        }

        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical order
        for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }

    // helper class
    private static class WordCount {
        String word;
        int count;
        WordCount(String w, int c) { word = w; count = c; }
    }
}

```

```

C:\hadoop-3.3.0\sbin>jps
11072 DataNode
20528 Jps
5620 ResourceManager
15532 NodeManager
6140 NameNode

C:\hadoop-3.3.0\sbin>hdfs dfs -mkdir /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /
Found 1 items
drwxr-xr-x - Anusree supergroup 0 2021-05-08 19:46 /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -copyFromLocal C:\input.txt /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r-- 1 Anusree supergroup 36 2021-05-08 19:48 /input_dir/input.txt

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input.txt
hello
world
hello
hadoop
bye

```

```

C:\hadoop-3.3.0\sbin>hadoop jar C:\sort.jar samples.topn.TopN /input_dir/input.txt /output_dir
2021-05-08 19:54:54,582 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-08 19:54:55,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1620483374279_0001
2021-05-08 19:54:55,821 INFO input.FileInputFormat: Total input files to process : 1
2021-05-08 19:54:56,261 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1620483374279_0001
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-08 19:54:56,843 INFO conf.Configuration: resource-types.xml not found
2021-05-08 19:54:56,843 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-08 19:54:57,387 INFO impl.YarnClientImpl: Submitted application application_1620483374279_0001
2021-05-08 19:54:57,507 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ES0:8088/proxy/application_1620483374279_0001/
2021-05-08 19:54:57,508 INFO mapreduce.Job: Running job: job_1620483374279_0001
2021-05-08 19:55:13,792 INFO mapreduce.Job: Job job_1620483374279_0001 running in uber mode : false
2021-05-08 19:55:13,794 INFO mapreduce.Job: map 0% reduce 0%
2021-05-08 19:55:20,020 INFO mapreduce.Job: map 100% reduce 0%
2021-05-08 19:55:27,116 INFO mapreduce.Job: map 100% reduce 100%
2021-05-08 19:55:33,199 INFO mapreduce.Job: Job job_1620483374279_0001 completed successfully
2021-05-08 19:55:33,334 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=65
FILE: Number of bytes written=530397
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=142
HDFS: Number of bytes written=31
HDFS: Number of read operations=8
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

```

```

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
hello 2
hadoop 1
world 1
bye 1

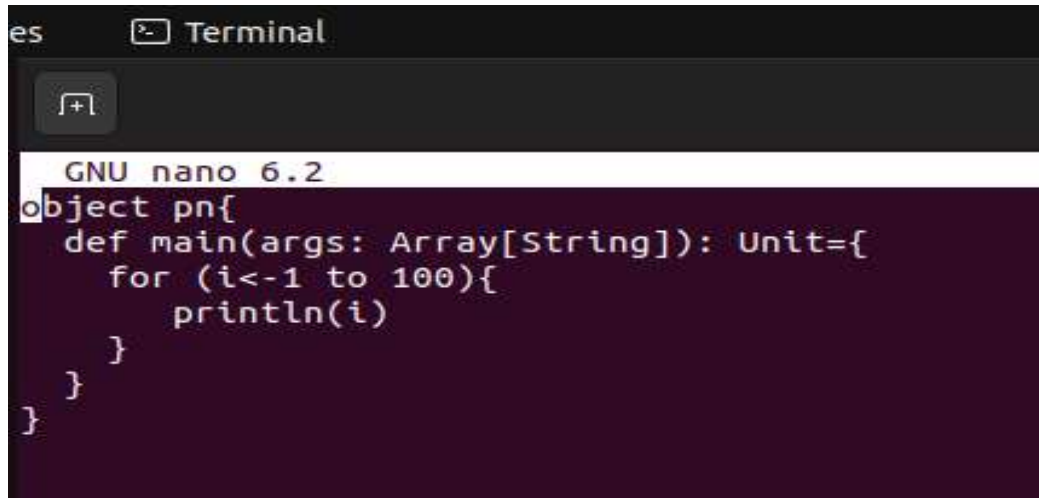
C:\hadoop-3.3.0\sbin>

```

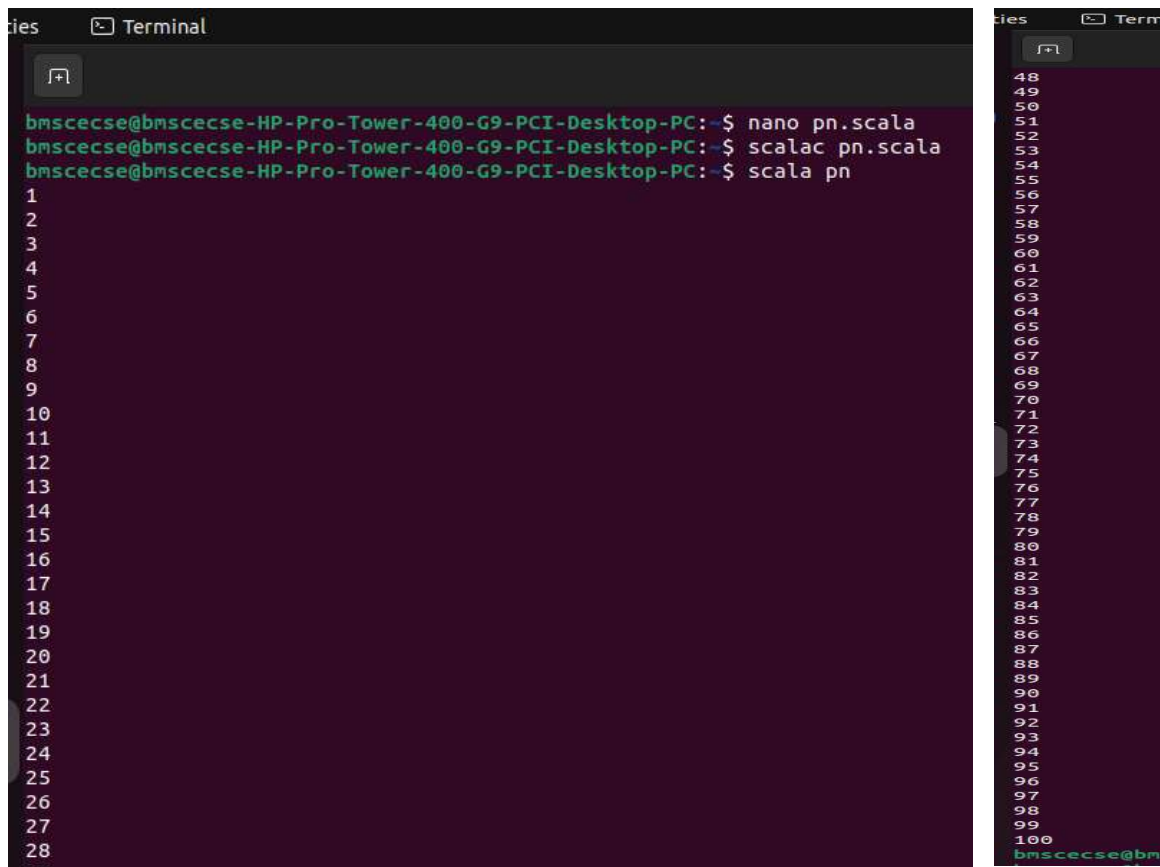

LABORATORY PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT



```
es Terminal
GNU nano 6.2
object pn{
  def main(args: Array[String]): Unit={
    for (i<-1 to 100){
      println(i)
    }
  }
}
```



```
ies Terminal
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ nano pn.scala
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ scalac pn.scala
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ scala pn
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
48
49
50
51
52
53
54
55
56
57
58
59
60
61
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63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
bmscecse@bm
```

LABORATORY PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

CODE, COMMAND WITH OUTPUT

```
bmscscse@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: $ spark-shell
26/05/04 16:13:42 WARN Utils: Your hostname, bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC resolves to a loopback address: 127.0.1.1; using 10.124.3
26/05/04 16:13:42 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/04 16:13:48 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.124.3.47:4040
Spark context available as 'sc' (master = local[*], app id = local-1777891430182).
Spark session available as 'spark'.
Welcome to

  ____      _
 / ___|    / \
| |  | |  / _ \
| |  | | / ___ \
| |  | |/_/   \_\
| |  | |
|_|  |_|      _/
              \_\

version 3.5.4

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.26)
Type in expressions to have them evaluated.
Type :help for more information.

scala> val textFile = sc.textFile("/home/bmscscse/wc.txt")
textFile: org.apache.spark.rdd.RDD[String] = /home/bmscscse/wc.txt MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = textFile.flatMap(line => line.split(" ")).map(word => (word,1)).reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:23

scala> import scala.collection.immutable.ListMap
import scala.collection.immutable.ListMap

scala> val sorted = ListMap(counts.collect().sortBy(_._2 > _._2): _*)
sorted: scala.collection.immutable.ListMap[String,Int] = ListMap(hello -> 6, spark -> 6, big -> 2, data -> 2, world -> 2)

scala> for((k,v) <- sorted) { if(v > 4) println(k + ", " + v) }
hello,6
spark,6

scala>
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))
```

```
df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered"))
```

```
# Step 5: Explode the lemmatized words and show results
```

```
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...  
[nltk_data] Unzipping corpora/stopwords.zip.  
[nltk_data] Downloading package wordnet to /root/nltk_data...  
+-----+  
|word |  
+-----+  
|hello |  
|hate  |  
|hate  |  
|love  |  
|dont  |  
|want  |  
|cant  |  
|put   |  
|nobody|  
|else  |  
+-----+
```